
Climate Canary

Raumüberwachung mit Fokus auf
Datenschutz, Zuverlässigkeit und
Benutzerfreundlichkeit

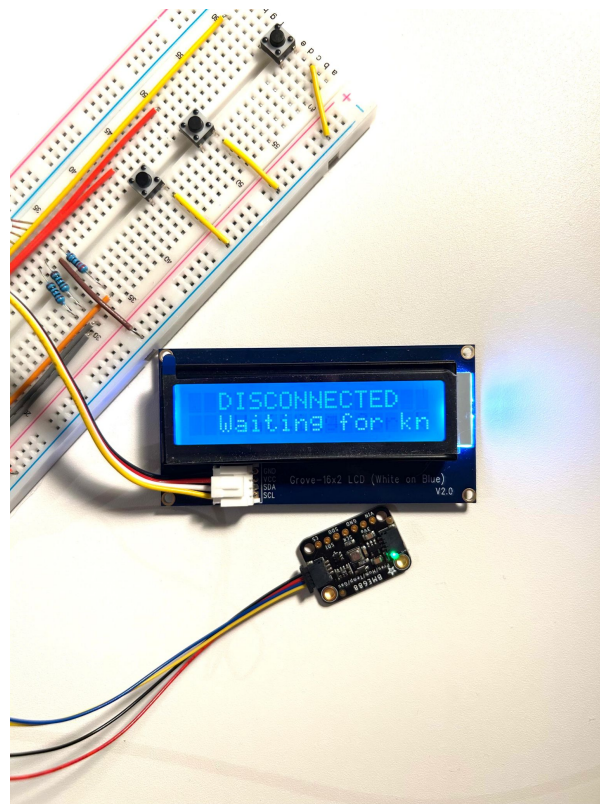
PS Software Engineering SS/2026

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Zuverlässigkeit und Ausfallsicherheit

- Benutzerfreundliches Setup
- LED Zustände mit Farben & Blinkmustern
- Live-Status direkt am Arduino Display
- Detailliertes Logging am Raspberry PI
- Kein Datenverlust bei Verbindungsproblemen:
 - Bis zu 1000 Messwerte lokal gespeichert
 - Automatische Wiederübertragung
 - 48h Retry-Mechanismus zwischen RPI und Backend





Rollenbasierte Zugriffslogik

- **EMPLOYEE** → eigenes Büro & common areas
- **DEP LEAD** → Department Overview
- **MANAGEMENT** → anonymisierte Aggregationen
- **BUILDING ADMIN** → Vollzugriff

Room-Fair Aggregation

- Räume werden zuerst einzeln aggregiert
- Danach erst Gesamttaggregation
- Verhindert unfairen Einfluss einzelner Räume

Bucket-Logik – resolveBucketSize()

≤ 7 Tage → “raw”

Keine Aggregation, jeder Messwert wird einzeln mit timestamps zurückgegeben

≤ 90 Tage → “1h”

Messwerte werden stündlich gemittelt

≤ 1 Jahr → 6h

Messwerte werden in 6h Blöcken gemittelt

default → 1d

Messwerte werden in 6h Blöcken gemittelt

Department Lead: ≤ 30 Tage → 1d, default 1w
(forceReduced())

Frontend und User Experience

Room Climate Data

Building administration view for persisted measurements, trends, and threshold violations.

Room ↑	Type ↑↓	Department ↑↓
Mgmt Office A	Office	Management
Mgmt Office B	Office	Management
Room 1	Office	Research (2)
Sales Common Area	Common area	Sales
Sales Office A	Office	Sales

<< < 1 2 > >>

Mgmt Office A
Main Office - Management

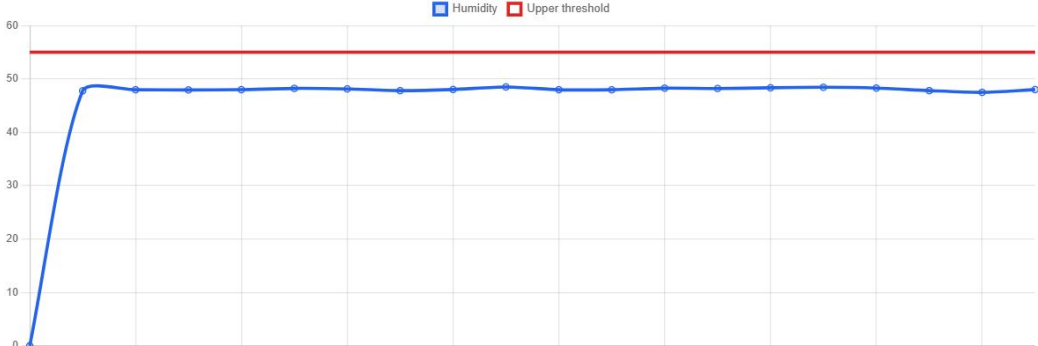
Humidity

24 hours (raw) < Today >

25.05.26 - 25.05.26

Jump to 25.05.2026

Bucket: raw



Time (hours)	Humidity	Upper threshold
0	0	55
1	48	55
2	48	55
3	48	55
4	48	55
5	48	55
6	48	55
7	48	55
8	48	55
9	48	55
10	48	55
11	48	55
12	48	55
13	48	55
14	48	55
15	48	55
16	48	55
17	48	55
18	48	55
19	48	55
20	48	55
21	48	55
22	48	55
23	48	55
24	48	55

Frontend und User Experience

Welcome, Iris

Department climate changes and warnings

 5
Departments

 6
Warnings

 5
Departments affected

 Engineering

2 warnings

Week and month compared with previous equal period

Humidity: 1

Air Quality: 1

STATISTIC	LAST WEEK	LAST MONTH
Humidity	Improved	Unchanged
Temperature	Unchanged	Unchanged
Pressure	Improved	Unchanged
Air Quality	Worsened	Worsened

 HR

1 warning

Week and month compared with previous equal period

Humidity: 1

STATISTIC	LAST WEEK	LAST MONTH
Humidity	No data	No data
Temperature	No data	No data
Pressure	No data	No data
Air Quality	No data	No data

 Management

1 warning

Week and month compared with previous equal period

Temperature: 1

STATISTIC	LAST WEEK	LAST MONTH
Humidity	No data	No data
Temperature	No data	No data
Pressure	No data	No data
Air Quality	No data	No data

Frontend und User Experience



Filter by Employee

All employees

Absence Types

Holiday

Sick Leave

Parental Leave

Other

Month

Week

Agenda

<

Today

>

May 2026

Jump to

25.05.2026

Holiday

Sick Leave

Parental Leave

Other

Mon	Tue	Wed	Thu	Fri	Sat	Sun
27	28	29	30	01	02	03
04	05	06	07	08	09	10
Ben Schmidt — Holiday						
11	12	13	14	15	16	17

Team Absences

Engineering

Manage Absences

Absence Overview

Employee

All employees

Status

All statuses

From

Start

To

End

4 absences

1 pending approval

Ben Schmidt

Holiday

PERIOD

05/05/2026 – 09/05/2026

Planned

Approve

Reject

Ben Schmidt

Sick Leave

PERIOD

03/02/2026 – 04/02/2026

Approved

Clara Huber

Holiday

PERIOD

10/02/2026 – 14/02/2026

Approved

Frontend und User Experience

 Dashboard Absences Building 2 Logout @eva Eva Kowalski

Threatolds Climate Hints

+ Add Threshold

Eng Office A 4 thresholds

HUMIDITY 1 threshold

Type ↑	Limit Value ↑↓	Enabled	Actions
LOWER	65	☒	Edit Delete

TEMPERATURE 2 thresholds

Type ↑	Limit Value ↑↓	Enabled	Actions
LOWER	18	☒	Edit Delete
UPPER	27	☒	Edit Delete



Frontend und User Experience

The screenshot displays a web application for managing sensor stations. A modal window titled "Scan for Arduino Devices" is open, providing instructions: "The Raspberry Pi will scan for nearby Arduino devices in setup mode (takes ~10 seconds). Select a found device to register it as a sensor station." The modal includes a "Start Scan" button and a table with the following data:

BLE MAC Address	Action
7C:DE:CE:44:CC:B1	<button>✓ Select</button>

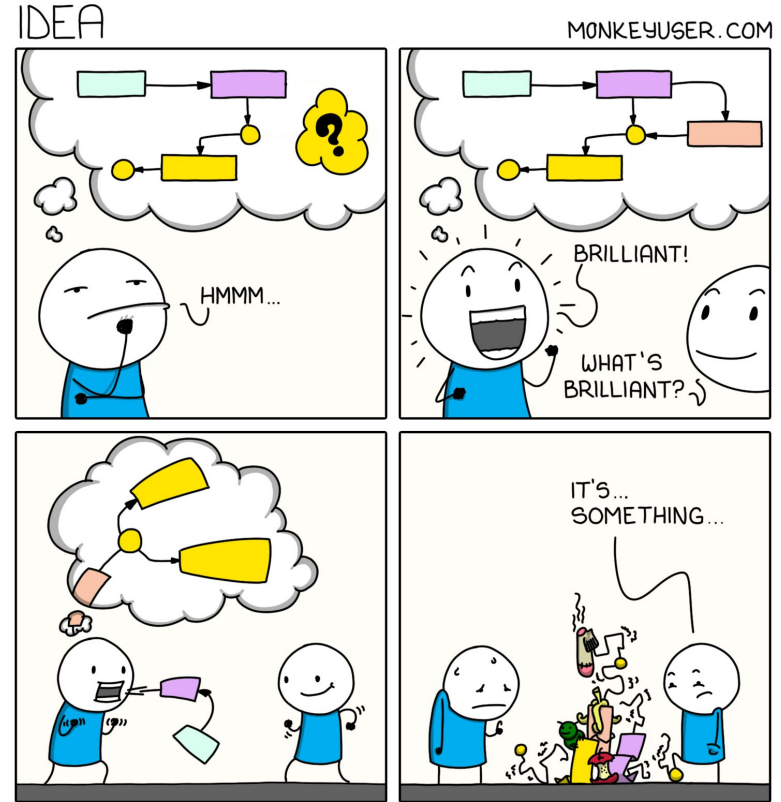
The modal also features a "Close" button. In the background, the main interface shows a "Sensor Stations — rpi-test25" section with a "Scan for Arduinos" button. Below this, a table lists sensor stations:

ID	Name	Status
17	7C:DE:CE:44:CC:B1	Needs Setup

Other visible elements include a "Download conf.yaml" button, a "Host Name" field with the value "rpi-test25", and a "Status" field with the value "ONLINE".

Entwicklung & Teamarbeit

- Agiles Projektmanagement
- Modularer Systemaufbau
- Inkrementeller Entwicklungsprozess
- Kontinuierliche Lauffähigkeit des Systems
- Selbstorganisation und Eigeninitiative im Team
- Echtzeit-Kommunikation über Instant Messaging
- Gemeinsame Ausarbeitung von Konzept und Systemdesign





Live Demo

